SECTION 1 IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY

1.1 Product identifiers

Product Form	Substance
Trade Name	Carboxymethyl Cellulose (CMC) Micron Powder for Li-ion Battery Anode Materials
Product Number	NG08BE0309
CAS Number	9004-32-4

1.2 Relevant identified uses of the substance or mixture and uses adviced against

Use of the substance/mixture	Research

1.3 Details of the supplier of safety data sheet

Company	Nanografi Nanotechnology
Address	ODTÜ Teknokent İkizler Binası B-1/H ODTÜ Teknokent 06531- ANKARA
Phone	+90 312 285 85 09
Fax	+90 312 210 13 09

1.4 Emergency Telephone Number

Emergency Telephone Number	+90 312 285 85 09

SECTION 2 HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

Not a hazardous substance or mixture according to Regulation (EC) No. 1272/2008.

2.2 Label Elements

Hazard Pictograms	None
Signal Word	None

Hazard Statements	H402 Harmful to aquatic life. H412 Harmful to aquatic life with long lasting effects.
Precautionary Statements	P273 Avoid release to the environment. P501 Dispose of contents/ container to an approved waste disposal plant.

2.3 Other Hazards

None.

SECTION 3 COMPOSITION/INFORMATION ON INGREDIENTS

3.1 Substances

Synonyms	Carboxy methyl cellulose sodium, Carboxymethyl cellulose, Carboxymethyl cellulose sodium salt, CMC
CAS Number	9004-32-4

3.2 Mixtures

Not applicable.

SECTION 4 FIRST AID MEASURES

4.1 Description of first aid measures

General Advice	Consult a physician. Show this safety data sheet to the doctor in attendance.
First-aid measures after inhalation	Provide fresh air.
First-aid measures after skin contact	Wash off with soap of water. If irritation persists consult a physician.
First-aid measures after eye contact	Rinse with plenty of water, also under the eyelids. If symptoms persist, call a physician.

First-aid measures after ingestion	Clean mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Consult a physician if symptoms develop.

4.2 Most important symptoms and effects, including acute and delayed

None.

4.3 Indication of any immediate medical attention and special treatment needed

None.

4.4 Other Information

No data available.

SECTION 5 FIREFIGHTING MEASURES

5.1 Extinguishing media

Suitable Extinguishing Media	Water. Carbon dioxide (CO2). Dry powder.

5.2 Special hazards arising from the substance or mixture

Specific hazards during fire-fighting	Carbon Oxides, Sodium Oxides.

5.3 Advice for firefighters

Special protective equipment for firefighters	Wear self-contained breathing apparatus.

5.4 Further Information

No data available.

SECTION 6 ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Personal Precautions	Ensure adequate ventilation. Wear personal protective equipment. Avoid dust formation. Avoid breathing vapors, mist or gas.

6.2 Environmental precautions

Prevent further leakage or spillage if safe to do so. Do not let product enter drains. Discharge into the environment must be avoided.

6.3 Methods and material for containment and cleaning up

Use mechanical handling equipment. Avoid dust formation. Pick up and transfer to properly labelled containers. Adequate disposal.

6.4 Reference to other sections

For further information refer to section 13.

SECTION 7 HANDLING AND STORAGE

7.1 Precautions for safe handling

Precautions for Safe Handling	Avoid formation of dust and aerosols. Provide appropriate ventilation at places where dust is formed. Wear personal protective equipment.
Advice on Protection Against Fire and Explosion	Normal measures for preventive fire protection.

7.2 Conditions for safe storage, including any incompatibilities

Storage Conditions	Keep containers tightly closed in a dry and well-ventilated place.

7.3 Specific end use(s)

Apart from the uses mentioned in section 1.2 no other specific uses are stipulated.

SECTION 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

Contains no substances with occupational exposure limit values.

8.2 Exposure controls

Appropriate Engineering Controls	Handle in accordance with good industrial hygiene and safety practice. Wash hands before breaks and at the end of the workday.
Hand Protection	Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands.
Eye Protection	Eye protection (EN 166).
Skin and Body Protection	Protective work clothing. Wear close-toed shoes and long sleeves/pants.
Respiratory Protection	No protective equipment is needed under normal use conditions.
Protective Measures	Facilities storing or utilizing this material should be equipped with an eyewash facility, safety shower, and ventilation systems. Handle in accordance with good industrial hygiene and safety practice.
Environmental Exposure Controls	Prevent further leakage or spillage if safe to do so. Do not let product enter drains.

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Physical State	Solid
Appearance	Powder
Color	White
Odor	No data available
Odor Threshold	No data available
pH	6.5-8.5
Relative Evaporation Rate	No data available
Melting Point	No data available
Freezing Point	No data available
Boiling Point	No data available
Flash Point	No data available
Auto-ignition Temperature	No data available
Decomposition Temperature	No data available
Flammability (solid, gas)	No data available
Vapor Pressure	No data available
Relative vapor density at 20°C	No data available
Relative Density	No data available
Solubility in H2O	Soluble
Log Pow	No data available
Viscosity, Kinematic	No data available
Viscosity, Dynamic	No data available
Explosive Properties	No data available
Oxidizing Properties	No data available
Explosive Properties	No data available

9.2 Other information

No other information available.

SECTION 10 STABILITY AND REACTIVITY

10.1 Reactivity

No data available.

10.2 Chemical stability

Stable under recommended storage conditions.

10.3 Possibility of hazardous reactions

No data available.

10.4 Conditions to avoid

No data available.

10.5 Incompatible materials

No data available.

10.6 Hazardous decomposition products

No data available.

SECTION 11 TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Acute Toxicity	Not classified.
Skin corrosion/irritation	Not classified.
Serious eye damage/irritation	Not classified.
Respiratory or Skin Sensitization	Not classified.
Germ Cell Mutagenicity	Not classified.
Carcinogenity	No component of this product present at levels greater than or equal to 0.1% is identified as probable, possible or confirmed human carcinogen by IARC.

	No component of this product present at levels greater than or equal to 0.1% is identified as a known or anticipated carcinogen by NTP. No component of this product present at levels greater than or equal to 0.1% is identified as a carcinogen or potential carcinogen by OSHA.
Reproductive Toxicity	Not classified.
STOT-Single Exposure	Not classified.
STOT-Repeated Exposure	Not classified.
Aspiration Hazard	Not classified.
Other Information	To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.

SECTION 12 ECOLOGICAL INFORMATION

12.1 Toxicity

Ecology-general	No data available.
Acute aquatic toxicity	Harmful to aquatic life. LC50 - Oncorhynchus mykiss (rainbow trout) - 100 - 1,000 mg/l - 96h EC50 - Daphnia - 87.26 mg/l 48h
Chronic aquatic toxicity	Harmful to aquatic life with long lasting effects.

12.2 Persistance and degradability

No data available.

12.3 Bioaccumulative potential

No data available.

12.4 Mobility in soil

No data available.

12.5 Results of PBT and vPvB assessment

No data available.

12.6 Other adverse effects

No data available.

SECTION 13 DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Product	Offer surplus and non-recyclable solutions to a licensed disposal company. Contact a licensed professional waste disposal service to dispose of this material.
Contaminated Packaging	Dispose of as unused product.

SECTION 14 TRANSPORT INFORMATION

Safe to carry by any means of transportation. Not restricted any mode of transportation.

SECTION 15 REGULATORY INFORMATION

15.1 Safety, health and environmental regulations

SARA 302 Components

No chemicals in this material are subject to the reporting requirements of SARA Title III, Section 302.

SARA 313 Components

This material does not contain any chemical components with known CAS numbers that exceed the threshold reporting levels established by SARA Title III, Section 313.

Massachusetts Right to Know Components

No components are subject to the Massachusetts Right to Know Act.

California Prop. 65 Components

This product does not contain any chemicals known to State of California to cause cancer, birth defects, or any other reproductive harm.

SECTION 16 OTHER INFORMATION

16.1 Abbreviation of acronyms

ADR	European Agreement concerning the International Carriage of Dangerous Goods by Road
CLP	Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008
IATA	International Air Transport Association
REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals Regulation (EC) No 1907/2006
PBT	Persistent Bioaccumulative Toxic
RID	Regulations concerning the International Carriage of Dangerous Goods by Rail
vPvB	Very Persistent and Very Bioaccumulative

16.2 Full text of H-statements

H402	Harmful to aquatic life.
H412	Harmful to aquatic life with long lasting effects.

16.3 Hazard classification

Health	0
Flammability	0
Reactivity	0
Special Hazards	

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.